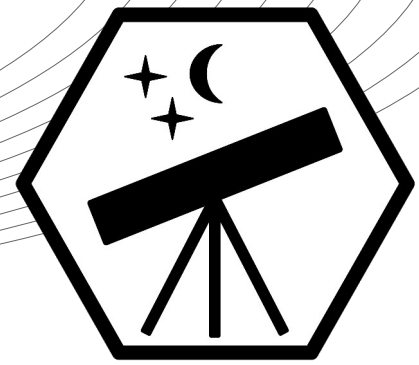


International Astronomy and Astrophysics Competition

Pre-Final Round 2025



Important: Read all the information on this page carefully!

General Information

- The six problems are separated into three categories: 2x basic problems (A; 4 points), 2x advanced problems (B; 6 points), 2x research problems (C; 8 points). The research problems require you to read a short scientific article to answer the questions. A link to the PDF article is provided.
- You can write your solution in the blank space under the problems or type your solution digitally on a computer. If you need more space for your solution, you can use extra sheets of paper.
- You receive points for the correct solution as well as for the performed steps. That means: You will not get full points for a correct value if the calculations and steps are missing.
- Make sure to **clearly** mark your final solution values (e.g., by underlining, red color, box).
- You can reach up to 36 points in total. You qualify for the Final Round if you reach at least 16 Points (Junior), 20 points (Youth), or 24 points (Senior).
- It is not allowed to work in groups. Assistance from teachers, friends, family, or the internet is prohibited. Textbooks and calculators are allowed. Cheating will result in immediate disqualification!

Uploading Your Solution

- Please upload a PDF document containing clear pictures of the problem sheet pages with your written solutions in the blank space, or upload a digitally typed document.
- You can upload your solution online via your account: <https://iaac.space/login>
- Only upload **one single PDF file!** If you have multiple pictures, please merge them into one file. Do not upload your solution in any different format. (Example: No Word or Zip files.)
- The submission deadline is **Sunday, 17 August 2025, 23:59 UTC+0.**
- The results of the Pre-Final Round will be announced on Monday, 1 September 2025.

Good luck!

Problem A.1: Scales of Gravitation (4 Points)

Every object around us exerts a gravitational force: from small items nearby to massive celestial bodies in our solar system and beyond. In this problem, you will compare the gravitational effects of various objects. For this comparison, assume your mass is 70 kg.

(a) Determine the gravitational force (in Newton [N]) between you and the following objects:

	Distance	Mass	Force
(1) Another Person	1 m	70 kg	ANSWER : $F \approx 3.7 \times 10^{-7} \text{ N}$
(2) Truck on the Street	3 m	30,000 kg	ANSWER : $F \approx 1.56 \times 10^{-5} \text{ N}$
(3) Moon	385,000 km	$7.3 \times 10^{22} \text{ kg}$	ANSWER : $F \approx 2.30 \times 10^{-3} \text{ N}$
(4) Jupiter	$780 \times 10^6 \text{ km}$	$1.9 \times 10^{27} \text{ kg}$	ANSWER : $F \approx 1.46 \times 10^{-5} \text{ N}$
(5) Alpha Centauri A	4.37 light-years	$1.1 M_{\odot}$	ANSWER : $F \approx 6.01 \times 10^{-12} \text{ N}$
(6) Andromeda Galaxy	$2.5 \times 10^6 \text{ light-years}$	$1.5 \times 10^{12} M_{\odot}$	ANSWER : $F \approx 2.51 \times 10^{-11} \text{ N}$

Notes: Express results in exponential format (e.g., $2 \times 10^{-3} \text{ N}$). One solar mass $M_{\odot} = 2 \times 10^{30} \text{ kg}$.

(b) Rank the objects from strongest to weakest gravitational effect on you.

strongest (3), (4), (2), (1), (6), (5) **weakest**

(c) Which objects have a very similar force despite being of much different scale?

Truck on the Street: 1.56×10^{-5} and Jupiter: 1.46×10^{-5}

WORKINGS OF A.1 (a):

A.1 Another Person

$F = G \frac{m_1 m_2}{r^2}$

F = Gravitational Force (N)
 m_1, m_2 = Two masses (kg)
 r = distance between two centers (m)
 G = Gravitational constant ($6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$)

1 km = 10^3 m
 1 light year = $9.461 \times 10^{15} \text{ m}$
 1 solar mass $M_{\odot} = 2 \times 10^{30} \text{ kg}$

ROW 1
 My Mass = 70 kg (m_1)
 Another Person's mass = 70 kg (m_2)
 Distance (r) = 1 m

$F = G \frac{m_1 m_2}{r^2}$

$= (6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2) \cdot \frac{(70 \text{ kg})(70 \text{ kg})}{(1 \text{ m})^2}$

CALCULATOR: $6.674 \times \text{EXP } -11 \times 70 \times 70 \div (1 \times 1)$

ANSWER: $F \approx 3.7 \times 10^{-7} \text{ N}$

ROW 2 Truck on the Street

$6.674 \times 10^{-11} \times \frac{70 \text{ kg} \times 30,000 \text{ kg}}{(3)^2}$

$6.674 \times 10^{-11} \cdot 70 \text{ kg} \cdot 30,000 \text{ kg} \div (3 \times 3)$

ANSWER: $F \approx 1.56 \times 10^{-5} \text{ N}$

ROW 4 Jupiter

$780 \times 10^6 \text{ km} \times 1000 = 7.80 \times 10^{11} \text{ m}$

$6.674 \times 10^{-11} \cdot 70 \cdot 1.9 \times 10^{27} = 8.87642 \times 10^{18}$

$(7.80 \times 10^{11})^2 = 6.084 \times 10^{23}$

$\frac{8.87642 \times 10^{18}}{6.084 \times 10^{23}} = 0.000014589$

ANSWER: $F \approx 1.46 \times 10^{-5} \text{ N}$

ROW 5 Alpha Centauri A

1 light year = $9.461 \times 10^{15} \text{ m}$
 $4.37 \text{ light years} \cdot 9.461 \times 10^{15} \text{ m} = 4.134457 \times 10^{16} \text{ m}$
 $1.1 M_{\odot} = 1.1 \cdot (2 \times 10^{30} \text{ kg}) = 2.2 \times 10^{30} \text{ kg}$

$6.674 \times 10^{-11} \cdot 70 \cdot 2.2 \times 10^{30} = 1.027796 \times 10^{22}$

$(4.134457 \times 10^{16})^2 = 1.71 \times 10^{33}$

$\frac{1.027796 \times 10^{22}}{1.71 \times 10^{33}} = 5.99441 \times 10^{-12}$

ANSWER: $F \approx 6.01 \times 10^{-12} \text{ N}$

ROW 3 Moon

$385,000 \text{ km} \times 1000 = 385,000,000 \text{ m} = 3.85 \times 10^8 \text{ m}$

$F = G \frac{m_1 m_2}{r^2} \rightarrow F = 6.674 \times 10^{-11} \frac{(70 \text{ kg})(7.3 \times 10^{22} \text{ kg})}{(3.85 \times 10^8 \text{ m})^2}$

$6.674 \times 10^{-11} \cdot 70 \cdot 7.3 \times 10^{22} = 3.410414 \times 10^{13}$

$(3.85 \times 10^8)^2 = 1.48225 \times 10^{17} \text{ m}$

$\frac{3.410414 \times 10^{13}}{1.48225 \times 10^{17}} = 0.002300835$

ANSWER: $F \approx 2.30 \times 10^{-3} \text{ N}$

ROW 6 Andromeda Galaxy

$1.5 \times 10^6 \cdot 2 \times 10^{30} = 3.0 \times 10^{36} \text{ kg}$

1 light year $\approx 9.461 \times 10^{15} \text{ m}$
 $2.5 \times 10^6 \cdot 9.461 \times 10^{15} = 2.36525 \times 10^{22}$

$6.674 \times 10^{-11} \cdot 70 \cdot 3.0 \times 10^{36} = 1.40154 \times 10^{24}$

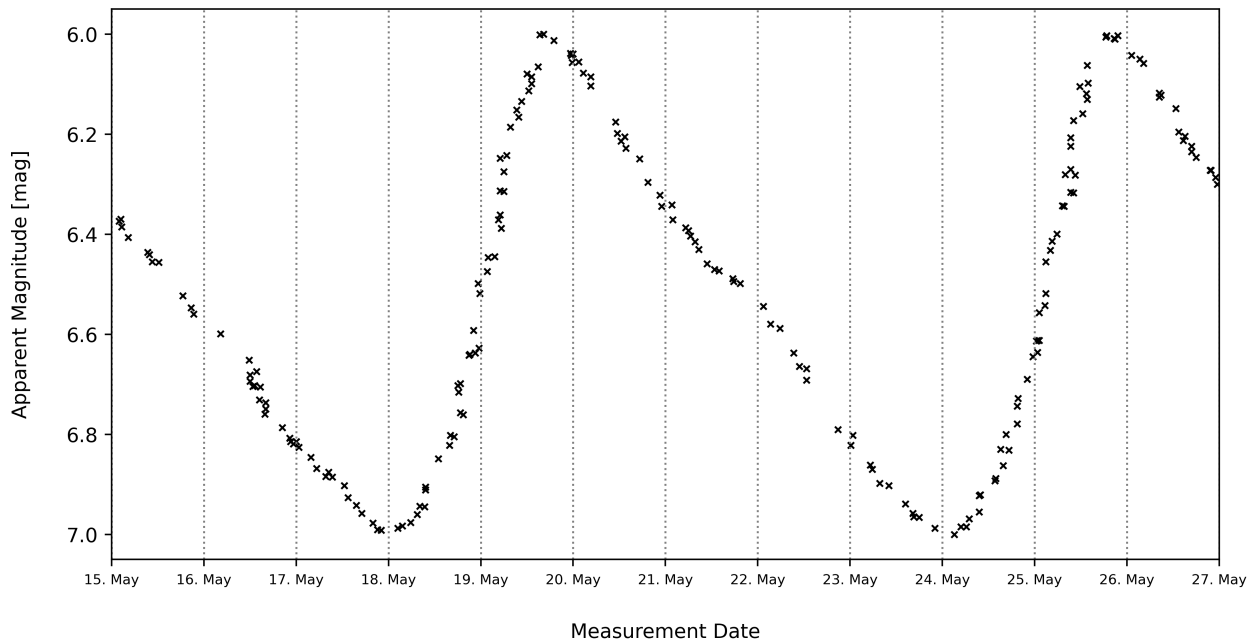
$(2.36525 \times 10^{22})^2 \approx 5.59441 \times 10^{44}$

$\frac{1.40154 \times 10^{24}}{5.59441 \times 10^{44}} \approx 2.50525 \times 10^{-11}$

ANSWER: $F \approx 2.51 \times 10^{-11} \text{ N}$

Problem A.2: Variable Star (4 Points)

In May 2025, a research team observed a section of the night sky continuously for several days. One star appears to change its brightness. Here is a plot of its apparent magnitude:



Stars with such a behaviour are called *Cepheids* and their absolute magnitude M can be calculated by applying the *Leavitt Law* (*Period–Luminosity relation*)

$$M = -2.43 \cdot (\log_{10}(T) - 1) - 4.05,$$

where T is the star's pulsation period in days. The research team finds that the star's average apparent magnitude is 6.5 mag. Determine the star's distance to the Earth in parsecs.

A.2
Leavitt Law

Absolute Magnitude:

$$\log_{10}(6) \approx 0.77815$$

$$\log_{10}(6) - 1 \approx -0.22185$$

$$M = -2.43 \cdot (-0.22185) - 4.05 \approx 0.5393 - 4.05 = -3.51$$

Distance:

$$m - M = 6.5 - (-3.51) = 10.01$$

$$10.01 + 5 = 15.01$$

$$\log_{10}(d) = \frac{15.01}{5} = 3.002$$

$$d = 10^{3.002} \approx 1005 \text{ parsecs}$$

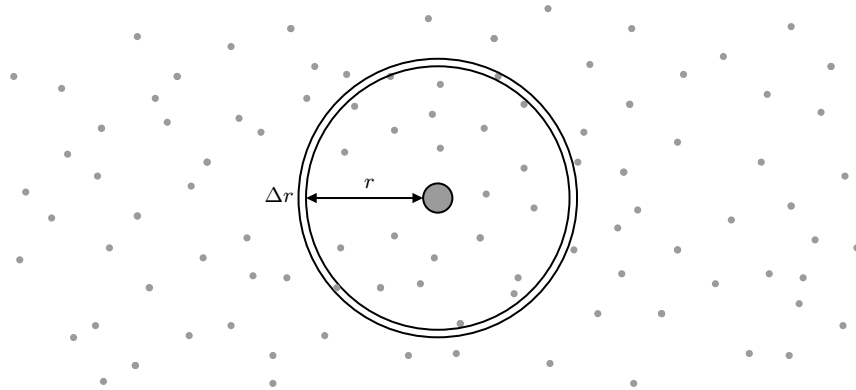
$$d = 1.005 \times 10^3 \text{ parsecs}$$

ANSWER: $1.005 \times 10^3 \text{ parsecs}$

Problem B.1: Brightness of the Night Sky (6 Points)

The night sky is not entirely dark because of the thousands of stars that shine. Some stars are closer to us, while others are farther away, which, along with other factors, affects their brightness. For simplicity, we assume that stars are uniformly distributed with a density of 0.004 stars per cubic light-year, and that their luminosity is comparable to that of the Sun (3.8×10^{26} watts).

Consider a thin spherical shell of radius r and thickness Δr with the Earth at its center:



- Find an expression for the number of stars $\Delta N(r)$ in this shell and for the total radiation power $\Delta P(r)$ reaching the Earth from this shell.
- Find an expression for the total power $P(r)$ received from all stars up to a distance r .
- The universe is infinite. What is the total power received for $r \rightarrow \infty$?
- Explain, why modern cosmology is relevant for correctly interpreting this result.

The result that the total power is infinite assumes an infinite static universe with uniform star density. Modern cosmology reveals that the universe is expanding, and its geometry affects light. The universe has a horizon beyond which light hasn't had time to arrive since the Big Bang (about 13.8 billion years ago). Cosmological models, like the Friedmann equations, account for these effects, showing the actual power received is finite due to the universe's age and expansion, not infinite as a simple static model suggests.

(a)

$$\Delta V = 4\pi r^2 \Delta r$$

$$\Delta N(r) = 0.004 \cdot 4\pi r^2 \Delta r$$

$$\Delta N(r) = 0.016\pi r^2 \Delta r \text{ stars}$$

$$\frac{L}{4\pi r^2} \quad \Delta P(r) = \Delta N(r) \cdot \frac{L}{4\pi r^2}$$

$$\Delta P(r) = (0.016\pi r^2 \Delta r) \cdot \frac{3.8 \times 10^{26}}{4\pi r^2}$$

$$\downarrow$$

$$\Delta P(r) = 0.016 \Delta r \cdot \frac{3.8 \times 10^{26}}{4}$$

$$\frac{3.8 \times 10^{26}}{4} = 0.95 \times 10^{26} = 9.5 \times 10^{25}$$

ANSWER: $\Delta P(r) = 0.016 \cdot 9.5 \times 10^{25} \Delta r$
 $\Delta P(r) = 1.52 \times 10^{24} \Delta r \text{ watts}$

(b)

$$P(r) = \int_0^r \Delta P(r') dr'$$

$$P(r) = \int_0^r 1.52 \times 10^{24} dr'$$

ANSWER: $P(r) = 1.52 \times 10^{24} \cdot r \text{ watts}$

B.1(c)

(c) $r \rightarrow \infty$ (If universe is infinite)
 $P(r) \rightarrow \infty$ (Then Power received is)
 $\downarrow$
ANSWER: $P(\infty) = \infty$ watts

Problem B.2: Non-relativistic Friedmann Equation (6 Points)

Besides other questions, cosmology studies how the universe will evolve in the future. To determine its fate, we can consider the non-relativistic form of Hubble's law,

$$v(t) = H(t) \cdot r(t),$$

which gives the recessional velocity v at a distance r from an observer at a given time t . It is possible to make predictions about the universe using non-relativistic mechanics. For this, we assume a constant matter density ρ throughout the universe and consider the behaviour of a sphere of radius r centered on an observer, containing all the matter within it.

(a) Find an expression for the kinetic energy E_{kin} and the potential energy E_{pot} of the sphere.

(b) Show that the total energy E is given by

$$E = \frac{4}{3}\pi G m r^2 (\rho_c - \rho),$$

where G is the gravitational constant, and find the expression for the *critical density* ρ_c .

(c) Discuss the fate of the universe for the following three scenarios:

$$\rho < \rho_c \qquad \rho = \rho_c \qquad \rho > \rho_c$$

(d) Derive the first non-relativistic Friedmann equation (with some constant k):

$$H^2 = \frac{8\pi G \rho}{3} - \frac{k}{r^2}$$

(a)

B.2 (a)

Hubble's Law: $v(t) = H(t) \cdot r(t)$

$$E_{kin} = \frac{1}{2} m v^2$$

$$E_{kin} = \frac{1}{2} m (Hr)^2 = \frac{1}{2} m H^2 r^2$$

$$m = \rho \cdot \frac{4}{3} \pi r^3$$

$$E_{kin} = \frac{1}{2} \left(\rho \cdot \frac{4}{3} \pi r^3 \right) H^2 r^2$$

$$E_{kin} = \frac{2}{3} \pi \rho H^2 r^5$$

$$E_{pot} = -\frac{3}{5} \frac{G m^2}{r}$$

$$m^2 = \left(\rho \cdot \frac{4}{3} \pi r^3 \right)^2 = \rho^2 \cdot \left(\frac{16}{9} \pi^2 r^6 \right) = \rho^2 \cdot \frac{16}{9} \pi^2 r^6$$

ANSWER:

$$E_{pot} = -\frac{3}{5} \frac{G \cdot \rho^2 \cdot \frac{16}{9} \pi^2 r^6}{r}$$

$$E_{pot} = -\frac{3}{5} \cdot \frac{16}{9} \pi^2 G \rho^2 r^5$$

$$E_{pot} = -\frac{48}{45} \pi^2 G \rho^2 r^5$$

(b)

B.2 (b)

$$E = E_{\text{kin}} + E_{\text{pot}}$$

$$E = \frac{2}{3} \pi \rho H^2 r^5 - \frac{4\pi}{3} G \rho^2 r^5$$

$$E = \frac{4}{3} \pi G \left(\rho \cdot \frac{4}{3} \pi r^3 \right) r^2 (\rho_c - \rho)$$

$$E = \frac{4}{3} \pi G \cdot \frac{4}{3} \pi \rho r^5 (\rho_c - \rho)$$

$$E = \frac{16}{9} \pi^2 G \rho r^5 (\rho_c - \rho)$$

$$0 = \frac{2}{3} \pi \rho H^2 r^5 - \frac{4\pi}{3} G \rho^2 r^5$$

$$H^2 = \frac{8\pi G \rho}{3} \rightarrow (\rho = \rho_c)$$

$$\rho_c = \frac{3H^2}{8\pi G}$$

$$E = -\frac{3}{5} \frac{G \left(\rho \cdot \frac{4}{3} \pi r^3 \right)^2}{r} \left(1 - \frac{\rho}{\rho_c} \right)$$

ANSWER: $\rho_c = \frac{3H^2}{8\pi G}$

(c)

$\rho < \rho_c$: density is less critical, $E > 0$, the universe expands infinite.

$\rho = \rho_c$: density=critical, $E=0$. The universe is constantly expanding.

$\rho > \rho_c$: density exceeds critical, $E < 0$. The universe may eventually collapse, due to gravitational attraction, (negative total-energy).

(d)

B.2 (d)

$$\frac{E}{m} = \frac{1}{2} H^2 r^2 - \frac{3}{5} \frac{Gm}{r}$$

$$\left(\rho \cdot \frac{3}{4} \pi r^3 \right)$$

ANSWER:

$$H^2 = \frac{8\pi G \rho}{3} - \frac{k}{r^2}$$

Problem C.1 : Milky Way-Andromeda Collision (8 Points)

This problem requires you to read the following recently published scientific article:

No certainty of a Milky Way-Andromeda collision.

Sawala, T., Delhomelle, J., Deason, A.J. et al. Nat Astron (2025).

Link: <https://www.nature.com/articles/s41550-025-02563-1.pdf>

Answer the following questions related to this article:

(a) Which objects in the Local Group and which physical processes are most relevant for determining collision probabilities?

MW, M31, M33, LMC. Gravitational interaction, dynamic friction, tidal effects

(b) Explain the meaning of each column presented in Table 1.

MW: Milky Way, M31 is Andromeda Galaxy, M33 is Triangulum, LMC is Large Magellanic Cloud, SMC is Small Magellanic Cloud.

(c) What is the difference between Figure 1 and Figure 2?

Figure 1, shows probability densities of MW and M31 positions in 100 Monte Carlo samples of the fiducial model, integrated over 10 Gyr or until the merger. While, Figure 2 displays 100 realizations of our fiducial Monte Carlo model and also state the probability for a MW-M31 merger within 10 Gyr.

(d) How do M33 and the LMC each affect the probability of a MW–M31 collision?

M33 increases merger probability, it reduces M31's transverse velocity relative to MW. LMC decreases merger probability, and its orbit is perpendicular to MW-M31, this increases MW's velocity and also adds perpendicular momentum.

(e) How likely is a merger occurring within 5 Gyr, assuming a 20 kpc distance threshold?

Approximately 20% to 30%

(f) What is gravit. softening, and why is a too small or too large softening length problematic?

A technique in N-body simulations that modifies gravitational force at small distances. If too small, it can cause two-body scattering and disrupt galaxy structures. If too large, could potentially miss collisions due to underestimating merger dynamics and close interactions.

Problem C.2 : New Solar System Objects (8 Points)

This problem requires you to read the following recently published scientific article:

Discovery and dynamics of a Sedna-like object with a perihelion of 66 au.

Chen, YT., Lykawka, P.S., Huang, Y. et al. Nat Astron (2025).

Link: <https://www.nature.com/articles/s41550-025-02595-7.pdf>

Answer the following questions related to this article:

(a) What do the terms *trans-Neptunian object*, *perihelion distance*, and *semi-major axis* mean?

TNO (Trans-Neptunian Object): Astronomical objects in the solar system that orbit the sun at a greater distance than Neptune.

Perihelion distance: Minimum distance from the Sun of an objects elliptical orbit, to which it represents the closest approach point.

Semi-major axis: This is half the length of a major axis of an elliptical orbit, assists as a measure of an orbit's average distance or size from the Sun.

(b) What are Sedna-like objects, and why is Ammonite one of them?

Sedna like objects are subsets of Trans-Neptunian Objects (TNO), characterized by large semi-major axis ($a > 200$ au) as well as large perihelion distances ($q > 60$ au), this indicates orbits that are highly detached from Neptune's gravitational influence.

Ammonite (2023 KQ14) qualifies as a Sedna-like object, with a semi-major axis ($a = 251.9 \pm 0.3$ au) and perihelion distance ($q = 65.9 \pm 0.2$ au), which fits the criteria of $a > 200$ au and $q > 60$ au.

(c) Explain the *perihelion gap*, and why Ammonite is important for it.

The perihelion gap refers to an apparent under-populated region in the orbital distribution of TNO, with semi-major axis (a) between 150 and 600 au and perihelion distances (q).

Ammonite (2023 KQ14) with the distances and semi-major axis (a and q), is the first to occupy this gap, filling the previously unexplained void.

(d) How is Ammonite's orbit different from the other Senda-like TNOs?

Ammonite's orbit differs from the other Sedna-like TNO's (e.g. 2012 VP113, Sedna), due to its unaligned longitude of perihelion, which falls outside the clustered (Varpi), about 254° .

(e) How does Ammonite provide evidence for an important event about 4.2 billion years ago?

Ammonite provides the evidence through its orbital stability and characteristics, as Ammonite and other Sedna-like objects may have shared a primordial orbital clustering during the important event 4.2 billion years ago.

(f) What challenges did the scientists face during their observations?

Due to the objects great distance from the Sun, the apparent magnitudes appear to be faint, this makes the detection difficult for scientists observing through telescopes.